

THE STRUCTURE OF MATHEMATICAL REALITY

Phase II — Chapter 4

Formal Systems and the Architecture of Proof

Assignment

Self-Study Curriculum

Write a structured argument (not an essay, use numbered points) addressing the following question: “If Godel showed that no formal system captures all mathematical truth, how can we trust the mathematics we use?” Your argument should distinguish between the reliability of proofs *within* a system and the completeness of any particular system. It should also explain why Godel’s result is a feature (telling us something important about the nature of mathematical truth) rather than a defect.